ARYAMAN ARORA

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EDUCATION

Duke University *Aug 2024 – May 2028*
Major: *B.S. in Data Science* (Mathematics & Computer Science dual major)
GPA: 3.6/4.0
Coursework: Data Structures & Algorithms, Machine Learning, Linear Algebra, Advanced Probability, Discrete Math, Multivariable Calculus, Microeconomics, Programming with Python and Java, Advanced Algorithm Design (CS330), Real Analysis (MATH431).

WORK EXPERIENCE

Legal Enterprises of America | *Data Analytics Intern,* Chicago, IL *May 2026 – Aug 2026*
- Improved acquisition-screening efficiency, measured by 40% less review time per firm, by documenting deal-team needs in a PRD and developing decision-support software such as automated dashboards to prioritize law-firm targets and forecast settlements.
- Increased deal-sourcing volume by 35% by building a 70-attribute Python scoring model for 2,000 firms through a REST API.
- Reduced manual data prep time by 4 hrs per week by creating an SQL ETL pipeline to restructure law-firm case data (via wrangling).
- Created a Python settlement forecasting tool using multiple linear regression to predict firm performance based on case data.

Hickey Lab|*Machine Learning Researcher*, Duke University *Oct 2025 – May 2026*
- Developed AI research tools (via Huggingface) to optimize information retrieval and reduce language model computing resources.
- Built a quantization pipeline compressing LLM weights to 4 bits, cutting inference cost and memory footprint for LLM use.
- Designed a RAG agent spanning ingestion, retrieval, and response synthesis, using HuggingFace AI models and the OpenAI API.

Big Idea Technologies | *Data Analytics intern*, New York, NY *May 2025 – Jul 2025*
- Rebuilt client onboarding flows using Celonis process mining and SQL transformations, automating a Salesforce CRM pipeline.
- Standardized 3,000+ financial records by monitoring anomalies in system logs, reducing reconciliation effort by ~40%.
- Built a data model mapping each node in the KYC process, improving fraud detection coverage, and trained classification datasets.

Selco Foundation | *Software Engineer,* Bangalore, India *Jun 2023 – Aug 2023*
- Enabled assessment of 15 solar sites by building location-analysis and monitoring tools to support deployment decisions.
- Trained a Python/scikit-learn random forest model on geological and meteorological data to optimize solar installation deployment.
- Automated feature engineering and built Seaborn visualizations to compare locations that have a solar-energy potential for farmers.
- This assisted two rural farmers in the Gopalaiah region access solar farming tools (through Selco), aiding their livelihoods.

PROJECTS

***Chargeplan*, Selco Foundation extended project** 2026
- Developed a Python engine for EV charging, minimizing modeled energy costs subject to vehicle deadlines and shared power limits.
- Built a React/FastAPI application that converted vehicle requirements and electricity tariffs into charging schedules, readiness summaries, and an interactive visualization.
- Implemented scenario analysis to evaluate how electricity prices, charge capacity, and departure times affected fleet readiness.

Systematic Trading Engine, Duke FinTech Society Jan 2026
- Built ARIMA forecasting models (statsmodels) generating equity trading signals, executed via the Interactive Brokers TWS API.
- Validated with walk forward backtesting and position sizing, evaluated on RMSE, MAPE, precision/recall, and Sharpe ratio.

Multisport Analytics Platform Jan 2026
- Developing Baseline, a full stack application using React, Python, Django, and PostgreSQL to track recreational triathletes' progress.
- Designing FIT/CSV data imports and benchmarck comparisons across training blocks, with traceable performance insights

CORE SKILLS

Languages: Python, SQL, Bash/Shell, Java, React.
Tools & Methods: Pandas, PostgreSQL, Docker, Git, GitHub, PyTorch, AWS, Hugging Face Transformers, Agile, Jira.
Data Analysis: Tableau, Excel, Power BI, matplotlib, Powerpoint, Seaborn, Salesforce, Pandas, SQL, NumPy.

ADDITIONAL

Awards: Captain, UAE National debate team (2022-2024), Pearson Certificate of Academic excellence.
On campus: Duke Bass Connections researcher under Dr. Ibrahim Mohedas, Duke AI Machine Learning club.
Extra-curriculars: Ironman Triathlete (Northern California 2026), Duke Triathlon, Duke Club soccer.